DGCA CPL / ATPL STUDY NOTES • RADIO NAVIGATION • CH 14

✈ Chapter 14: Secondary Surveillance Radar

SSR — Mode A, Mode C & Mode S

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1. Introduction & Principle

Secondary Surveillance Radar (SSR) is a **secondary radar** system: instead of relying on echoes, the ground station interrogates airborne transponders which transmit their own coded replies. This gives much more information than primary radar.

📡 SSR Advantages Over Primary Radar

- Identity (callsign equivalent code)
- Altitude (Mode C, no need for R/T)
- Strong, reliable signal (transponder active transmitter, not a passive reflector)
- Works regardless of aircraft size or material (no 4th power law limitation)
- Not affected by weather, ground clutter or sea return

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- Mode S: Data link — downlinked aircraft data; uplinked ATC messages

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14 Secondary Surveillance Radar (SSR)

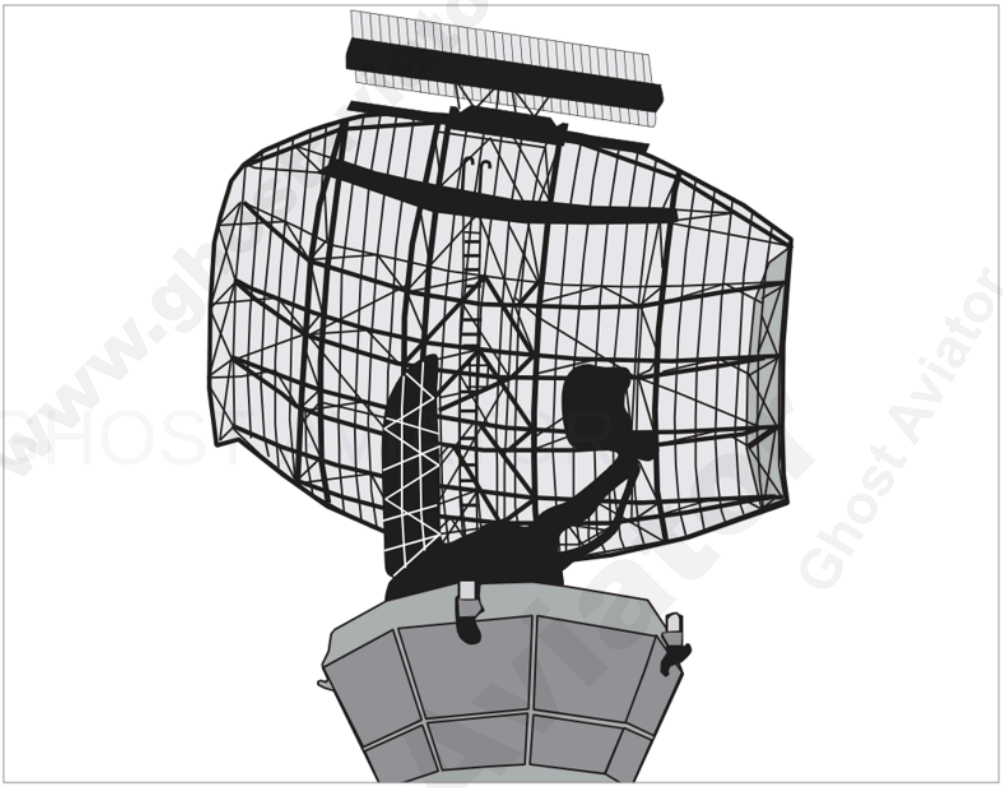


Figure 14.1 SSR aerial mounted on top of a 23 cm primary radar aerial

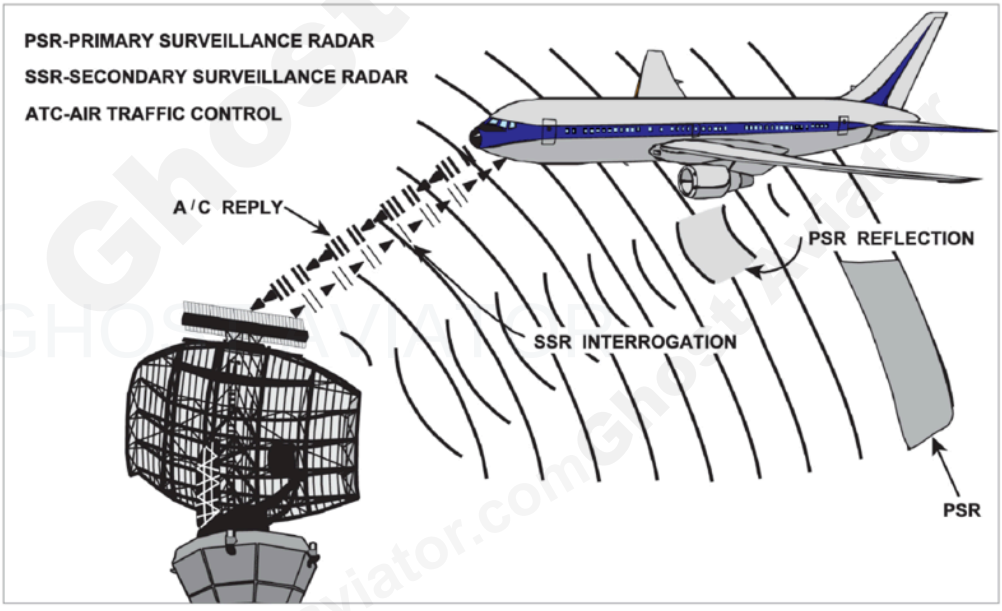


Figure 14.2 Primary and secondary radar used for ATC

Fig 14.1 & 14.2: SSR antenna (rotates with primary head) and overall principle of operation

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2. Ground Equipment

SSR Ground Station

Parameter	Detail
Transmit frequency	**1030 MHz** (UHF)
Receive frequency	**1090 MHz** (UHF)
Antenna type	Mounted on top of primary radar antenna; co-rotates
Antenna beam	Narrow in azimuth (1–3°)
SLS (Side Lobe Suppression)	Omnidirectional pulse P2 transmitted simultaneously with P1; if transponder sees P2 > P1, it suppresses reply → prevents side-lobe interrogations
Processor	Decodes replies, displays on PPI alongside primary return

3. Airborne Transponder

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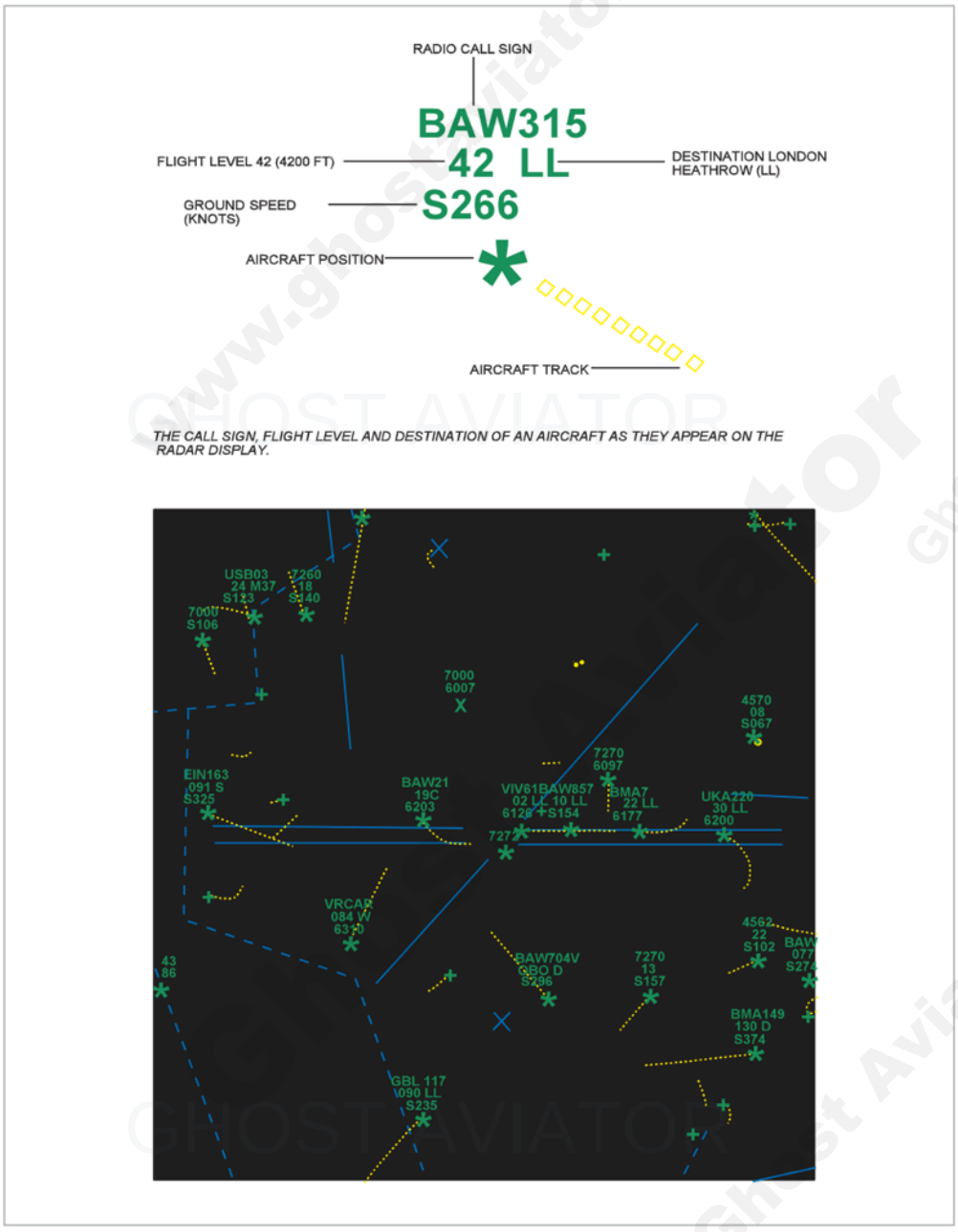


Figure 14.3 A radar display showing positions of aircraft in the London TMA

Fig 14.3: Transponder control panel — STBY, ON, ALT, IDENT positions, mode selector, code display

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Transponder Operation

Parameter	Detail
Receive frequency	**1030 MHz**
Transmit frequency	**1090 MHz**
Reply delay	**50 µs** (transponder receives and delays before replying, to allow distance measurement)
Antenna	Omni-directional (top and bottom fuselage)
STANDBY	System active, will NOT respond to interrogations (used on ground when not required)
ON	Mode A (identity) only
ALT	Mode C (altitude encoding) + Mode A
IDENT / SPI	Special Position Identification pulse — flashes radar blip for **20 seconds**

4. Interrogation Modes

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SSR Frequencies and Transmissions

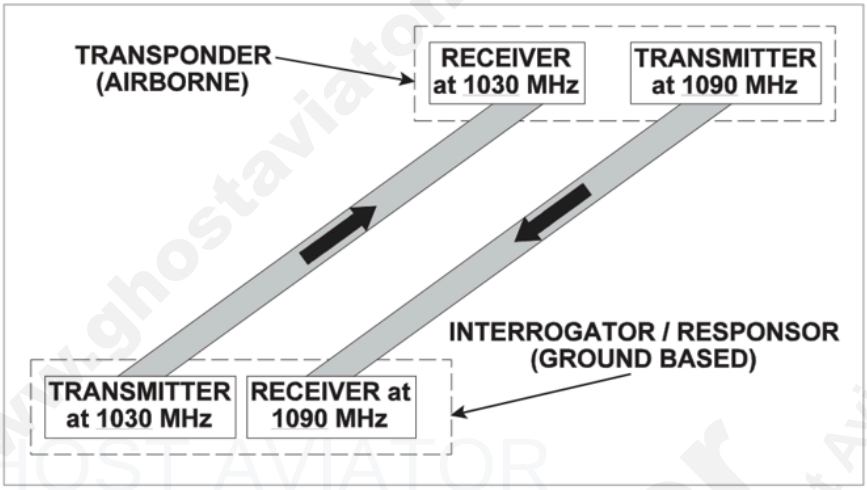


Figure 14.4 SSR operates in the UHF band

The ground station **transmits/interrogates** on **1030 MHz** and **receives** on **1090 MHz**. The aircraft **receives** on **1030 MHz** and **transmits/transponds** on **1090 MHz** after a delay of 50 microseconds. The SSR ground antenna transmits a narrow beam in the horizontal plane while the aircraft transmits omni-directionally i.e. the radiation pattern is circular around the aircraft.

Modes

The aircraft is interrogated from the ground station by a predetermined series of pulses on the carrier frequency of 1030 MHz; its transponder then transmits a coded reply on a carrier frequency of 1090 MHz. The two main modes of operation are:

- Mode A - an interrogation to identify an aircraft

Fig 14.4: Pulse spacings for Mode A (8 µs) and Mode C (21 µs) | Fig 14.5: Reply pulse train (F1...SPI...F2)

Mode Comparison

Mode	P1–P3 Spacing	Reply Contains	Info Provided
A (Identity)	**8 µs**	4096 identity codes	Aircraft identity (4-digit squawk)
B	17 µs	Identity codes	Aircraft identity (alternate)
C (Altitude)	**21 µs**	Altitude in 100 ft increments	Pressure altitude (re: 1013.25 hPa)
D	25 µs	Future use	—

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⚡ Key Mode Details

- Mode A provides **4096** identity codes (0000–7777 octal, with 1200, 7000, 7700 reserved/restricted)
- Mode C altitude: encoded in **100 ft** increments using Gillham code; referenced to **1013.25 hPa** regardless of QNH set
- Mode C displayed separately on PPI — ATC can read altitude without R/T
- Transponder replies with **framing pulses F1, F2** spaced 20.3 µs, with information pulses between

⚠ Required Actions

- When ATC say "SQUAWK IDENT": press IDENT button → SPI for 20 seconds
- HIJACK: squawk **7500**
- Loss of R/T: squawk **7600**
- Emergency: squawk **7700**
- Military conspicuity: squawk **7000**
- VFR conspicuity (no ATC unit): squawk **7000**

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5. Transponder Codes

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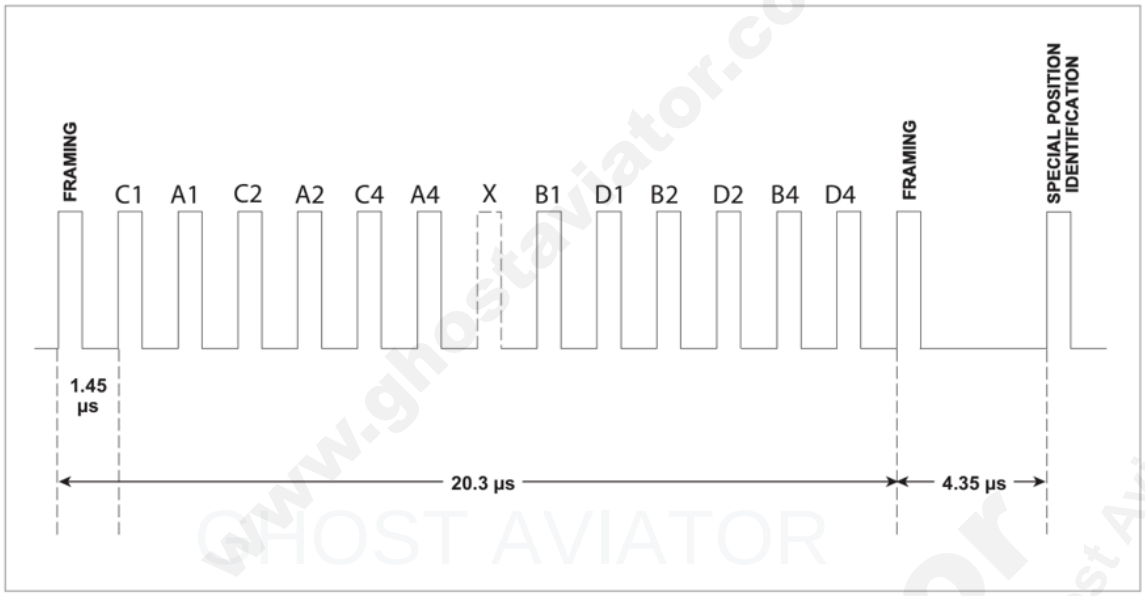


Figure 14.6 SSR reply pulse patterns

Fig 14.6: Reply codes — 4096 combinations of 4 digits 0-7 in Mode A

Code Summary

Code	Meaning
7700	Emergency
7600	Radio failure
7500	Unlawful interference (hijack)
7000	Military conspicuity / VFR conspicuity (UK)
2000	Entering controlled airspace without ATC clearance
Assigned	Four-digit code assigned by ATC

6. SSR Problems

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The aircraft transponder will reply correctly to a Mode A or C interrogation provided the pilot has correctly selected the mode and code allocated by ATC. On receiving a valid interrogation, the aircraft transponder transmits **two framing pulses**, F1 and F2, 20.3 µs apart. Between the framing pulses there are 12 usable **information pulses** (pulse X is for Mode B which is at present unused). A pulse can be transmitted or not, i.e. there are 2^{12} = 4096 possible combinations of pulses or codes which are numbered **0000 to 7777**; the figures 8 and 9 are not available.

A further pulse called the **Special Position Identification** (SPI) pulse may be transmitted together with the information pulses when the "Ident" button on the pilot's transponder is pressed, usually at ATC's request. This pulse is after the last framing pulse and will be automatically and continuously transmitted for about 20 seconds. It produces a distinctive display so that a controller can pick out a particular aircraft by asking the pilot to "**Squawk Ident**".

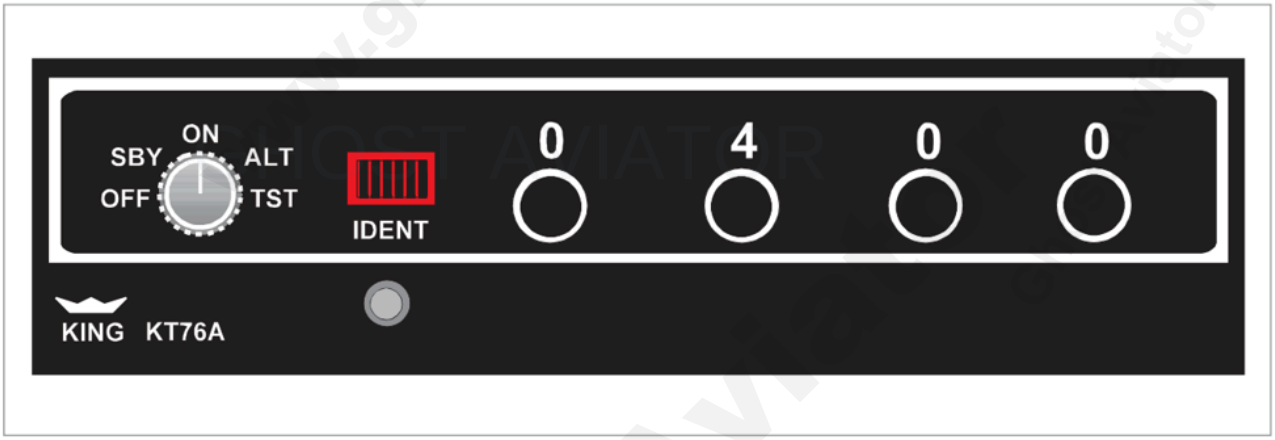


Figure 14.7 Transponder controls & indicators

Fig 14.7: Garbling (two aircraft within 1.7 NM cause overlapping replies) and Fruiting (random interrogations cause interference)

⚠ Garbling

- Two aircraft within **1.7 NM** of each other appear to be in the same direction from the ground station
- Their replies overlap → garbled, unreadable replies
- Solution:
  Mode S
  (selective addressing)

⚠ Fruiting (Phantoms)

- Multiple ground interrogators on same frequency
- Transponder receives interrogation from station B but reply is picked up by station A
- Station A sees a phantom target at incorrect position
- Solution:

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ISLS

(Improved Side Lobe Suppression) — synchronised interrogation timing between stations

7. Mode S

✓ Mode S Key Features

Feature	Detail
Addressing	Each aircraft has a unique **24-bit address** (16.7 million combinations)
Selective interrogation	Ground station interrogates ONLY the target aircraft → eliminates garbling
Data link	Bidirectional: aircraft can downlink data; ATC can uplink instructions
Altitude resolution	**25 ft** (vs 100 ft for Mode C)
ACAS / TCAS	Uses Mode S for aircraft-to-aircraft interrogation to provide collision avoidance
ELS	Elementary Surveillance — squawk, flight ID, altitude, emergency
EHS	Enhanced Surveillance — adds selected altitude, airspeed, heading (for upper airspace in Europe)
ADS-B	Automatic Dependent Surveillance Broadcast — squitter every 0.5 s; position via GPS

8. SSR Summary

Parameter	Detail
Type	Secondary radar; interrogation–reply
Ground Tx	1030 MHz (UHF)
Transponder Tx	1090 MHz (UHF)

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Transponder delay	50 µs
Mode A spacing	8 µs → 4096 codes → identity
Mode C spacing	21 µs → pressure altitude re 1013.25 hPa → 100 ft increments
SPI	IDENT pressed → 20 s enhanced blip
Garbling	<1.7 NM separation → overlapping replies
Fruiting	Phantom returns from multiple interrogators
Mode S address	24 bits → 16.7 million unique addresses
Mode S altitude	25 ft resolution

9. Practice Questions

Q1. SSR operates on frequencies:

(a) 1030 MHz ground Tx, 1090 MHz airborne Tx

(b) 1090 MHz ground Tx, 1030 MHz airborne Tx

(c) 9375 MHz both

(d) 960 MHz both

Answer: (a)
Ground transmits at **1030 MHz**, airborne transponder replies at **1090 MHz**.

Q2. Mode C provides ATC with:

(a) aircraft callsign

(b) airspeed

(c) position

(d) pressure altitude

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Answer: (d)

Mode C (P1–P3 spacing = 21 µs) provides **pressure altitude** referenced to 1013.25 hPa in 100 ft increments.

Q3. Advantage of Mode S over Mode A/C:

(a) no transponder required

(b) not dependent on pressure altimeter

(c) eliminates garbling by selective addressing

(d) uses VHF frequencies

Answer: (d)

Mode S eliminates garbling through **selective addressing** — each aircraft has a unique 24-bit ICAO address; ground station interrogates only one aircraft at a time.

Q4. A transponder code of 7700 indicates:

(a) radio failure

(b) emergency

(c) unlawful interference

(d) conspicuity

Answer: (b)

7700 = **Emergency**. 7600 = radio failure. 7500 = hijack (unlawful interference).

Q5. The unique identifier used in Mode S is:

(a) a 4-digit octal squawk

(b) the Mode C altitude code

(c) a 24-bit ICAO aircraft address

(d) a 12-bit reply pulse train

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Answer: (c)

Mode S uses a **24-bit ICAO address**, providing up to 16.7 million unique aircraft identifiers, enabling selective interrogation and data link capability.

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